

WHERE SUMMER SCHOOL



Number of frontal lectures hours: 27

Number of lab hours: 6

Credits (CFU): 6

Dynamics & Robotics:

- Newtonian dynamics in 3D,
- Lagrangian mechanics,
- Jacobians
- coordinate systems: gymnastics for expressing vectors in different frames.

Linear algebra:

- intuitive grasp of concepts from linear algebra
- understanding of linear ordinary differential equations, and a moderate level of mathematical maturity on linear systems.
- Eigenvalues, eigenvectors, rank, nullspace, and positive definite matrices. (to refresh Gilbert Strang's lectures on linear algebra)

Control Systems:

- State space representation of linear systems
- Stability criteria for systems of linear ordinary differential equations.

Jupyter Notebooks

- download **wheresim** project: <https://github.com/idra-lab/wheresim.git>
- follow install instructions in Readme.md

Locomotion code

- download **Quadruped-PyMPC**: https://github.com/iit-DLSLab/Quadruped-PyMPC/tree/feat/lesson_friday
- follow install instructions in Readme.md

Time Schedule




 Freie Universität Bozen
 Libera Università di Bolzano
 —
 Università Liedia de Bulsan



WHERE: Euregio Summer School — Timetable

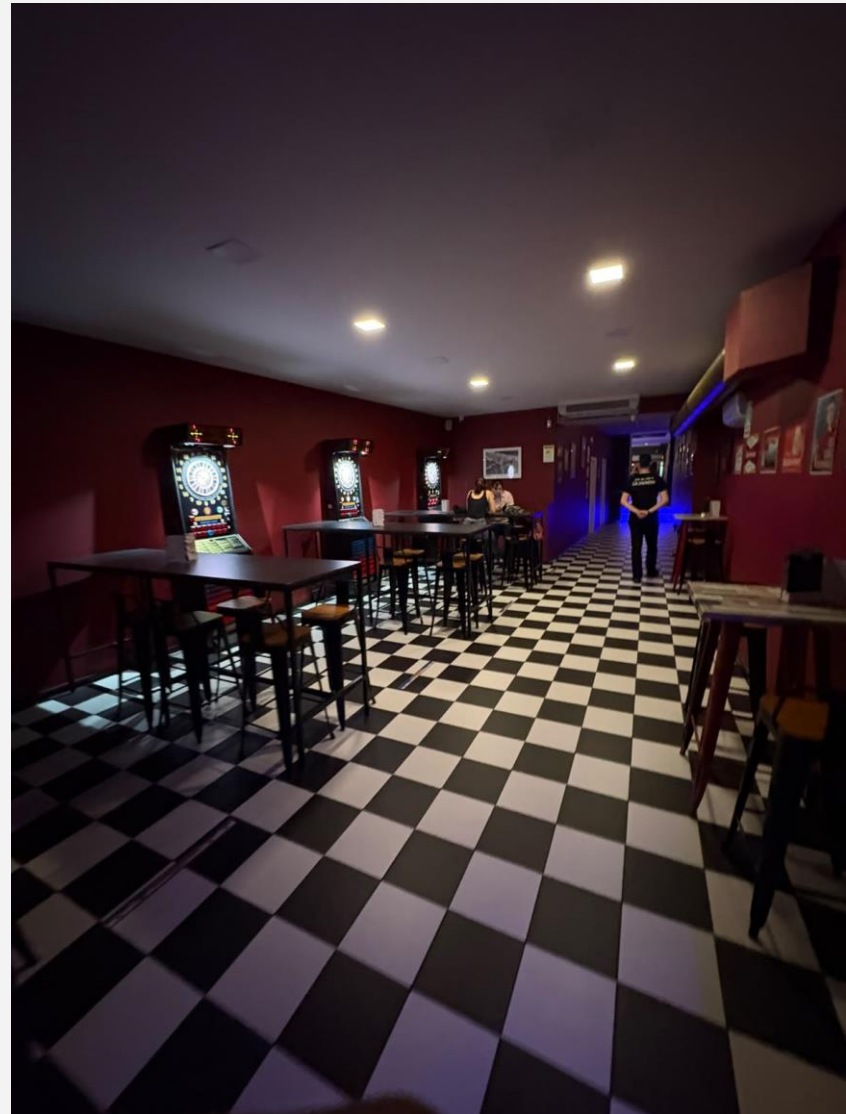
Time / Activity	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
09.00–10.40	Registration and welcome (9.00 - 10.00) Modeling of Legged Robots Instructor: Michele Focchi Location: Room Aula Garda	Primer on Optimization / Optimal Control Instructor: Marco Frego Location: Room Aula Garda	Trajectory Optimization Instructor: Enrico Mingo Location: Room Aula Garda	State Estimation for Legged Robots Instructor: Marco Camurri Location: Room Aula Garda	LABS: Locomotion Lab in Simulation Location: Room Aula Garda	Optional: Tutored Practical Session on Real Robot Location: Room Aula Garda
10.40–11.00	Coffee break	Coffee break	Coffee break	Coffee break	Coffee break	Coffee break
11.00–12.30	Modeling of Legged Robots Instructor: Michele Focchi Location: Room Aula Garda	Primer on Optimization / Optimal Control Instructor: Marco Frego Location: Room Aula Garda	Trajectory Optimization Instructor: Enrico Mingo Location: Room Aula Garda	State Estimation for Legged Robots Instructor: Marco Camurri Location: Room Aula Garda	LABS: Locomotion Lab in Simulation Location: Room Aula Garda	Optional: Tutored Practical Session on Real Robot Location: Room Aula Garda
12.30–14.00	Lunch	Lunch	Lunch	Lunch	Lunch	Lunch (not included)
14.00–15.40	Modeling of Legged Robots II Instructor: Michele Focchi Location: Room Aula Garda	Advanced Robot Control Instructor: Andrea Del Prete Location: Room Aula Garda	Sampling-based Control Instructor: Giulio Turrisi Location: Room Aula Garda	Reinforcement Learning I Instructor: Giulio Turrisi Location: Room Aula Garda	Reinforcement Learning II Instructor: Giulio Turrisi Location: Room Aula Garda	
15.40–16.00		Coffee break	Coffee break	Coffee break	Coffee break	
16.00–17.30		Advanced Robot Control Instructor: Andrea Del Prete Location: Room Aula Garda	Sampling-based Control Instructor: Giulio Turrisi Location: Room Aula Garda	Reinforcement Learning I Instructor: Giulio Turrisi Location: Room Aula Garda	Reinforcement Learning II Instructor: Giulio Turrisi Location: Room Aula Garda	
17.30–18.30	Poster Session + Coffe break					
18.30–20.00		Pub Session (not included)				
20.30–22.30				Social Dinner		

Pub session

Darts Pub

Via del Suffragio, 23, 38122 Trento

From 21:30 h onwards



Trekking in Dolomites



— Freie Universität Bozen
— Libera Università di Bolzano
— unibz
— Università Liedia de Bulsan



UNESCO heritage



Exam



For those that need credits

Multiple-choice exam

Done remotely after the school

Resources



- Introduction to Linear Algebra – Gilbert Strang (book) and course: <https://ocw.mit.edu/courses/mathematics/18-06-linear-algebra-spring-2010/>
- Feedback Control of Dynamic Systems - Gene F. Franklin, J. David Powell, Abbas Emami-Naeini
- A Mathematical Introduction to Robotic Manipulation - Richard M. Murray, Zexiang Li, S. Shankar Sastry.
- Introduction to Robotics: Mechanics and Control – J.Craig
- Kathib robotics video lectures:
http://videlectures.net/stanfordcs223aw08_introduction_robotics/
- Robotics Modelling, Planning and Control - Siciliano, B., Sciavicco, L., Villani, L., Oriolo, G.
- Springer Handbook of Robotics (Chapter 2, Dynamics) - Roy Featherstone and David Orin: https://link-springer-com.proxy.library.nd.edu/content/pdf/10.1007/978-3-540-30301-5_3.pdf
- Modern Robotics: Mechanics, Planning, and Control - Kevin M. Lynch and Frank C. Park: <http://hades.mech.northwestern.edu/images/7/7f/MR.pdf>
- Underactuated Robotics - Russ Tedrake:
<http://underactuated.csail.mit.edu/>

- ETH Robot Dynamics Lecture Notes:
https://ethz.ch/content/dam/ethz/special-interest/mavt/robotics-n-intelligent-systems/rsl-dam/documents/RobotDynamics2017/RD_HS2017script.pdf
- ETH ROS introduction course: <https://robohub.org/programming-for-robotics-introduction-to-ros/>
- Sebastien gros: Numerical optimization course:
<https://www.youtube.com/playlist?list=PLc2vvxBHfBcrzR8fhWc7qjT1lr51Kjue2>
- Convex Optimization - Stephen P. Boyd and Lieven Vandenberghe
https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf
- Numerical Optimization - Jorge Nocedal and Stephen Wright:
<https://link-springer-com.proxy.library.nd.edu/book/10.1007/978-0-387-40065-5>